| | | |
|---|---|---|
| | **GLOBAL ECONOMICS**<br>**Grade: 11th - Learning Evidence 3**<br>**Statistical models and research planning**<br>**Teacher: Nicolás López Cuéllar** | **1st TERM**<br>**2026–2027** |

| | |
|---|---|
| **Learning objective:** Apply continuous probability models and sampling distributions to investigate economic and financial questions, justify statistical methods, and communicate evidence-based conclusions. | **Assessment criteria:**<br>C8: Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges.<br>C9: Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample.<br>C10: Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions. |

**Name:** _____ **Grade:** _____ **Date:** _____

## Criteria assessment

Show all relevant calculations and clearly justify your answers. Partial credit may be awarded when the correct procedure is clearly demonstrated, even if arithmetic errors are made. No points will be awarded for answers without a demonstrated procedure, except for the brief research selection in Question 17.

A minimum of 4 points is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

| **Problems** | **C8** | **C9** | **C10** |
|---|---|---|---|
| 1–5: C8 direct (1 point each) | 5 | – | – |
| 6–10: C8 inverse (1 point each) | 5 | – | – |
| 11–13: C9 direct (2 points each) | – | 6 | – |
| 14–16: C9 inverse (2 points each) | – | 6 | – |
| 17: C10 research selection | – | – | 6 |
| **Total: 28 points** | **10** | **12** | **6** |

Answer on separate paper and show your reasoning. Use only the supplied normal and Student's $t$ reference sheets; no interpolation, CDF calculator, or inverse-CDF calculator is needed. C8 is worth **10 points**, C9 **12 points**, and C10 **6 points**: **28 points total**. At least **4 points per criterion** are required to pass that criterion. All calculations within one problem share its stated allocation, including both interval endpoints.

## Models and table conventions

Problems 1–10 use the supplied signed-deviation model $X = m + aT$, $T \sim t_\nu$. Problems 11–16 compare

$$X_T = m + aT, \quad T \sim t_\nu,$$

$$X_N = m + aZ, \quad Z \sim N(0,1), \quad \nu = n - 1.$$

Here $n$ is the number of sample observations, $m$ is the location (and mean for $\nu > 1$), and $a > 0$ is the scale. Use the stated *approximate standard deviation* directly as $a$, without adjustment. This is a teaching approximation:

$$\mathrm{SD}(X_T) = a\sqrt{\frac{\nu}{\nu-2}} \quad (\nu > 2),$$

$$\mathrm{SD}(X_N) = a.$$

The two models share a scale, not an actual variance. The approximation is coarser for small $\nu$ and improves as $\nu$ increases. Negative deviations mean below the contextual reference (early or under budget); positive deviations mean above it (late or over budget). Inverse intervals enclose modelled deviations.

Use the supplied year-local normal and Student's $t$ reference sheets. The printed $t$ columns are one-tail probabilities $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$. Report C8 probabilities as implied by these entries, and contextual cutoffs to two decimals.

The normal table gives four-decimal cumulative areas $\Phi(z) = P(Z \le z)$ for positive $z$. For direct normal probabilities, round each standardized boundary to two decimals *before* lookup. For inverse normal problems, select the closest printed CDF entry. At targets 0.0500 and 0.9500, magnitudes 1.64 and 1.65 tie; choose $|z| = 1.64$, as in the C9 activity. Use the CDF table for inverse normal lookups.

For **every C9 problem**, calculate both results and

$$E_A = |r_T - r_N|,$$

$$E_{Pt} = \frac{100E_A}{|r_T|}, \qquad E_{Pn} = \frac{100E_A}{|r_N|}.$$

For direct problems, $r_T, r_N$ are probabilities; for inverse problems, they are contextual cutoffs after back-transformation.

Compare inverse interval endpoints separately. A zero denominator makes that percentage undefined; the data below avoid zero denominators. These measures compare model results and do not establish error against observed truth. Report C9 probabilities and their $E_A$ to four decimals; contextual cutoffs and their $E_A$ to two decimals with units; and percentage comparisons to two decimals. Use unrounded back-transformed cutoffs for inverse comparisons, rounding only final results.

All results are table-based approximations, not exact distribution values: printed critical values and CDF entries are rounded. Both models are continuous, so endpoints have probability zero. The formally stated criteria above are retained; the supplied activity models govern the C8/C9 calculations here.

**1. Courier arrival deviations [C8-1]**

**C8 – Direct left tail.** A sample of $n = 10$ courier trips has arrival-time deviation from schedule, in minutes, with mean $m = 1$ and approximate standard deviation $a = 3$. Use $X = 1 + 3T$, $T \sim t_{n-1}$. Find $P(X < -4.499)$ and interpret the threshold. **[1]**

**2. Freight cost deviations [C8-1]**

**C8 – Direct right tail.** A sample of $n = 13$ consignments has cost deviation from budget, in dollars, with mean $m = -4$ and approximate standard deviation $a = 8$. Use $X = -4 + 8T$, $T \sim t_{n-1}$. Find $P(X > -10.984)$ using a complement, and explain why it exceeds 0.5. **[1]**

**3. Warehouse energy deviations [C8-1]**

**C8 – Direct interval above the mean.** Over $n = 17$ days, electricity-use deviation from forecast, in kWh, has mean $m = 2$ and approximate standard deviation $a = 6$. Use $X = 2 + 6T$, $T \sim t_{n-1}$. Find $P(7.19 < X < 12.476)$. **[1]**

**4. Crop shipment deviations [C8-1]**

**C8 – Direct interval below the mean.** A sample of $n = 8$ crop shipments has mass deviation from forecast, in tonnes, with mean $m = -1$ and approximate standard deviation $a = 3$. Use $X = -1 + 3T$, $T \sim t_{n-1}$. Find $P(-8.095 < X < -3.688)$. **[1]**

**5. Retail revenue alerts [C8-1]**

**C8 – Direct unequal combined tails.** Over $n = 7$ weeks, revenue deviation from plan, in thousands of dollars, has mean $m = 2$ and approximate standard deviation $a = 4$. Use $X = 2 + 4T$, $T \sim t_{n-1}$. Find the probability of an alert: $X < -5.772$ or $X > 7.76$. **[1]**

**6. Operating cost cutoff [C8-1]**

**C8 – Inverse lower-tail cutoff.** Over $n = 15$ production days, operating-cost deviation from budget, in dollars, has mean $m = -3$ and approximate standard deviation $a = 5$. Use $X = -3 + 5T$, $T \sim t_{n-1}$. Find $x$ such that $P(X < x) = 0.10$. **[1]**

**7. Loading time cutoff [C8-1]**

**C8 – Inverse upper-tail cutoff.** A sample of $n = 21$ loading jobs has duration deviation from target, in minutes, with mean $m = 2$ and approximate standard deviation $a = 7$. Use $X = 2 + 7T$, $T \sim t_{n-1}$. Find $x$ such that $P(X > x) = 0.025$. **[1]**

**8. Demand deviation interval [C8-1]**

**C8 – Inverse central interval.** Over $n = 12$ weeks, demand deviation from forecast, in tonnes, has mean $m = 4$ and approximate standard deviation $a = 9$. Use $X = 4 + 9T$, $T \sim t_{n-1}$. Find $L < U$ enclosing the central 90% of deviations with equal excluded tails. **[1]**

**9. Package mass alerts [C8-1]**

**C8 – Inverse equal-tail alert limits.** A sample of $n = 9$ packages has mass deviation from target, in grams, with mean $m = -0.4$ and approximate standard deviation $a = 1.2$. Use $X = -0.4 + 1.2T$, $T \sim t_{n-1}$. Find two alert limits leaving a combined outside probability of 0.05, split equally. **[1]**

**10. Dispatch time alerts [C8-1]**

**C8 – Inverse unequal-tail alert limits.** A sample of $n = 16$ dispatches has departure-time deviation from schedule, in minutes, with mean $m = -2$ and approximate standard deviation $a = 6$. Use $X = -2 + 6T$, $T \sim t_{n-1}$. Find $L, U$ with $P(X < L) = 0.01$ and $P(X > U) = 0.10$. **[1]**

**11. Sales revenue shortfalls [C9-2]**

**C9 – Direct left tail.** Over $n = 10$ weeks, sales-revenue deviation from target, in thousands of dollars, has mean $m = -2$ and approximate standard deviation $a = 5$. Compare $X_T = -2 + 5T$, $T \sim t_{n-1}$, with $X_N = -2 + 5Z$. Find $P(X < -11.165)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$, and interpret the comparison. **[2]**

**12. Delivery schedule deviations [C9-2]**

**C9 – Direct right tail.** A sample of $n = 15$ deliveries has arrival-time deviation from schedule, in minutes, with mean $m = 3$ and approximate standard deviation $a = 4$. Compare $X_T = 3 + 4T$, $T \sim t_{n-1}$, with $X_N = 3 + 4Z$. Find $P(X > -0.472)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$, and interpret the comparison. **[2]**

**13. Fuel forecast deviations [C9-2]**

**C9 – Direct interval crossing the mean.** Over $n = 13$ operating days, fuel-use deviation from forecast, in litres,

has mean $m = 1$ and approximate standard deviation $a = 6$. Compare $X_T = 1 + 6T$, $T \sim t_{n-1}$, with $X_N = 1 + 6Z$. Find $P(-7.136 < X < 14.074)$. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$, and interpret the comparison. **[2]**

### 14. Production cost savings [C9-2]

**C9 – Inverse lower-tail cutoff.** A sample of $n = 8$ batches has cost deviation from budget, in dollars, with mean $m = -1$ and approximate standard deviation $a = 2.5$. Compare $X_T = -1 + 2.5T$, $T \sim t_{n-1}$, with $X_N = -1 + 2.5Z$. Find the cutoff $x$ below which 5% of deviations fall. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$, and interpret the comparison. **[2]**

### 15. Order processing cutoff [C9-2]

**C9 – Inverse upper-tail cutoff.** A sample of $n = 17$ orders has processing-time deviation from target, in minutes, with mean $m = 2$ and approximate standard deviation $a = 3.5$. Compare $X_T = 2 + 3.5T$, $T \sim t_{n-1}$, with $X_N = 2 + 3.5Z$. Find the cutoff $x$ exceeded by 10% of deviations. Find both model results, calculate $E_A$, $E_{Pt}$, and $E_{Pn}$, and interpret the comparison. **[2]**

### 16. Cargo mass deviation interval [C9-2]

**C9 – Inverse central interval.** A sample of $n = 10$ cargo loads has mass deviation from target, in kg, with mean $m = 1.5$ and approximate standard deviation $a = 4$. Compare $X_T = 1.5 + 4T$, $T \sim t_{n-1}$, with $X_N = 1.5 + 4Z$. Find the central 95% interval with equal excluded tails under each model. Calculate $E_A$, $E_{Pt}$, and $E_{Pn}$ separately for each endpoint, and interpret. **[2]**

### 17. Choose your research option [C10-6]

**During the exam:** select **1** of the following options and write a very brief description (1–2 sentences) of real-life data that could plausibly follow your selected distribution. **[6]**

- **(a)** A continuous variable modelled by a linear probability density over a stated range.
- **(b)** Waiting times or intervals between events modelled by an exponential distribution.
- **(c)** Normally distributed measurements.

No datasets, sources, calculations, parameter estimates, or detailed model justification are required during the exam.

**Online report submission:** your PDF report will focus on the **1** selected option and must be submitted virtually through **Open**. This online submission is separate from the brief response written during the exam. To demonstrate the full C10 criterion, investigate an economic or financial question using real data and your selected continuous model. Include simulation, sampling analysis, and portfolio-risk measures; justify the selected methods and communicate results, assumptions, limitations, and evidence-based conclusions. The 6 points shown here assess the brief exam response; the report provides the broader C10 evidence. Select only one option and give a brief description (1–2 sentences) of plausible real-life data. **[6]**